

Project Status

Project Status — The Full Story, In Plain English

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This document explains the entire project so far — from what the disease is, to exactly what was researched each day, to what the numbers mean, to why the final answer leans positive — written so that someone with no medical or statistics background can follow every step. Wherever a technical term shows up, it's explained right there, the first time it's used.

If you only read one document to understand this project cold, read this one.

1. What is this project, in one paragraph?

A pharmaceutical company called **Amgen** makes a drug called **dazodalibep**. It's being tested as a treatment for an autoimmune disease called **Sjögren's disease**, where the body's own immune system mistakenly attacks the glands that make tears and saliva — causing severe dry eyes, dry mouth, fatigue, and joint/muscle pain. There is currently **no approved medicine that treats the actual disease** — only things that ease the dryness symptoms (eye drops, saliva substitutes). Amgen has finished (or is finishing) two large, final-stage clinical trials of dazodalibep. This project's job is to predict — using only publicly available evidence, before the company announces anything — whether those trials will come back as a genuine success or not, and to explain exactly why, the same way a stock market analyst predicts an earnings result before a company reports it.

2. The disease and the drug — why would this even work?

What's going wrong in the body

In Sjögren's disease, two types of immune cells — **T-cells** and **B-cells** — end up working together in a harmful way. Normally, T-cells “approve” B-cells to start producing antibodies (proteins that fight off real infections). In Sjögren's, this approval process goes into overdrive inside the tear and saliva glands, and the B-cells end up producing harmful antibodies that attack the person's own tissue. Over time, little clusters of this runaway immune activity build up inside the glands — almost like tiny, misplaced lymph nodes — and they keep fueling the disease.

The specific signal being blocked

The “approval” T-cells give B-cells happens through a specific molecular handshake: a protein called **CD40L** (found on the T-cell) plugs into a matching socket called **CD40** (found on the B-cell). Once that handshake happens, the B-cell gets the green light to multiply and start pumping out antibodies. Block that handshake, and the theory is you calm down the disease at its source, not just its symptoms.

Two ways exist to interrupt this handshake: - **Block the plug** (CD40L) — this is dazodalibep's approach. - **Block the socket** (CD40) — this is the approach of a competing drug called iscalimab, made by a different company (Novartis). Either approach, if it works, should produce the same outcome: no more “green light” reaching the B-cell.

Why dazodalibep is built the way it is

Dazodalibep is **not a normal antibody drug**. It's an engineered protein built on something called a **Tn3 scaffold** — think of it as a small, purpose-built molecular clip designed to grab onto CD40L, with no “antibody tail” (called an Fc region) attached to it. This detail matters a lot, for a specific historical reason:

In the early 2000s, a first generation of CD40L-blocking drugs was abandoned because they caused dangerous blood clots. The reason: those drugs *were* normal antibodies, and CD40L also happens to sit on the surface of blood-clotting cells (platelets). The antibody's tail could accidentally grab onto multiple platelets at once and stick them together, causing clots. Because dazodalibep has no antibody tail at all, it's specifically engineered to avoid triggering that exact mechanism. That doesn't mean zero risk of any blood clot ever (see the safety section below — one did occur in the small early trial), but it does mean the *specific, well-understood* mechanism that killed off the first generation of these drugs shouldn't apply the same way to this one.

3. Day 1 — What we researched, step by step, in plain terms

Day 1's whole job was **fact-finding**: gathering every real, sourced piece of evidence needed before making any kind of prediction. Nothing was predicted yet — that came in Day 2.

Step 1: Did the drug actually show any effect in its first, smaller test?

Before running the two big trials, Amgen ran a smaller early trial (called “Phase 2”) on about 180 patients total, split into two groups: - **Group 1** (“systemic” patients): people whose disease was affecting their whole body, not just dryness — joints, organs, blood tests, etc. - **Group 2** (“symptomatic” patients): people whose main problem was day-to-day symptoms (dryness, fatigue, pain) without much whole-body organ involvement.

Each group was randomly split again into “real drug” and “fake drug” (placebo) sub-groups, and neither the patients nor the doctors scoring them knew who got which — this is called a **double-blind, placebo-controlled** design, and it's the gold standard for proving a real effect, because it removes wishful thinking from the result.

What “the score went down” actually means: both groups were graded using scoring systems that work like a checklist — a higher number means worse disease, so an improvement means the number went *down*. - **ESSDAI** = a doctor-filled checklist scoring how active the disease looks across different organs (used for Group 1). - **ESSPRI** = a patient-filled questionnaire about pain, fatigue, and dryness (used for Group 2).

The actual results: - **Group 1 (ESSDAI):** the real-drug patients' score dropped by 6.3 points on average; the fake-drug patients' score dropped by 4.1 points. So real drug patients did about 2.2 points better than fake-drug patients. Statisticians ran the numbers and found there was only about a **1.7% chance** this 2.2-point gap happened purely by random luck (this “chance it's a fluke” number is called a **p-value** — smaller is better, and under 5% is the normal bar for “this is probably real”). - **Group 2 (ESSPRI):** real-drug patients improved by 1.8 points; fake-drug patients improved by only 0.5 points — a 1.3-point gap, with only a **0.02% chance** this was a fluke. This was actually the single cleanest, most convincing result in the whole early trial.

But not everything was clean. When researchers checked “how many patients hit a specific improvement target” (rather than just the average score), the real-drug group didn't clearly beat the fake-drug group by enough to rule out chance. This matters — it's an early hint of a bigger problem covered in Step 3 below.

Step 2: Blood tests that back up the story

Beyond the symptom scores, doctors also checked blood markers tied directly to the disease process: - A marker called **CXCL13** (a chemical signal that helps build those harmful immune clusters inside the glands) dropped

sharply in drug-treated patients, and bounced back up when the drug was stopped. - **Rheumatoid factor**, a type of harmful antibody, was also significantly reduced by the drug.

This matters because it's **independent proof the drug is doing exactly what the science predicted** — not just proof that a symptom score moved, which could theoretically happen for other reasons.

A safety note, stated precisely: one patient in the early trial developed a blood clot and a liver problem, and the trial doctors did officially link both to the drug. The important nuance: this happened 180 days *after* the last dose — well outside the window you'd expect if the mechanism were the same platelet-clumping problem that killed off the first generation of similar drugs (described in Section 2 above). One other patient in the trial developed a breast cancer diagnosis that doctors also linked to the drug, and one patient died of COVID-19 complications (considered unrelated to the drug). All of this needs to be reported honestly in a final deck — a small early trial with a handful of serious events like this doesn't prove or disprove anything on its own, but it shouldn't be glossed over either.

Step 3: Is the “ruler” used to measure success actually a good ruler?

This is one of the most important — and least obvious — things this project dug into. The ESSDAI scoring system (the doctor checklist) has a real, documented history of being **not very sensitive** — meaning even drugs that probably work have “failed” past trials using this exact scoring system, likely because: - Patients getting the **fake** treatment often improve anyway, just from natural ups-and-downs of the disease, or from the extra medical attention that comes with being in a trial. This is called the **placebo effect**, and it shrinks the gap between “real drug” and “fake drug” scores, making a real effect harder to prove. - Researchers even built an actual mathematical formula, based on many past trials, showing how much the fake-drug group tends to improve depending on how sick the patients were to begin with — the sicker the starting point, the bigger the fake improvement tends to be, purely because there's more room to naturally fluctuate downward.

This is a genuine risk baked into the disease and the scoring system itself, not something specific to Amgen's drug — and it means a “failed” result in a big trial doesn't necessarily mean the drug doesn't work; it could mean the ruler wasn't sensitive enough to detect a real, smaller effect.

Step 4: What happened when other companies tried this?

We researched every other drug tested for this disease over roughly the last 10 years, to build a track record. Two important, carefully-verified findings:

1. **The two drugs that block this exact same signal** (dazodalibep, blocking one side, and iscalimab, blocking the other side) **both technically achieved their own early-trial goals**. That's a modest, genuinely encouraging pattern — though with only two data points, it's a hint, not a proven law.
2. **A different competing drug (ianalumab, made by Novartis, using a totally different mechanism) already ran its own big final-stage trial and reported real, positive results about a year ago**. This is hugely useful, because it's actual, real-world proof that this kind of trial *can* succeed in this disease — it's not just theory. But — and this took careful double-checking to get right — **both times that drug “won,” the win was narrow, not a landslide** (explained with exact numbers in Day 2 below). That's an important, honest data point: even a real effect in this disease might not produce an obvious, easy win.

We also researched a second registered outcome measure for the smaller trial (explained in Step 5) and confirmed exactly how strict a statistical bar the original small trial used, and a few other technical facts corrected during a later review pass (documented throughout the individual research files, not repeated here in full).

Step 5: A closer look at the second, smaller trial

Amgen’s second big trial (the one testing day-to-day symptoms rather than whole-body disease activity) actually grades success using **two separate measurements**, not just one: the ESSPRI questionnaire mentioned above, plus a second tool called **DASPRI** — a daily symptom diary patients fill in repeatedly, rather than a one-time questionnaire at a doctor visit. This is a real, material detail: a diary tool has different risks (people might skip days, for example) than a one-time questionnaire, and nobody outside the company knows yet whether both tools need to show a positive result, or just one, for this trial to count as a win.

4. Day 2 — Turning the research into an actual prediction

Day 1 gathered evidence. Day 2’s job was to **do the math** and land on an actual call: will the big trial pass or fail?

The key question: how big does the effect need to be?

The big trial (NCT06104124, the systemic/whole-body one) has **651 patients** — about nine times more than the 74 patients in the small early trial’s matching group. More patients in a trial generally makes it **easier to prove a real effect exists**, even if that effect is smaller than what the early trial showed. The technical question is: exactly how much smaller an effect can this bigger trial still detect and call “real”?

To answer that, you need to know how much natural, random up-and-down “noise” to expect in patient scores. This is the single trickiest, most important judgment call in the whole model, and — importantly — **there isn’t one universally agreed-upon number for it**. We tested three different, reasonable estimates instead of just picking the most convenient one:

Which estimate	Where it comes from	How big an effect the trial needs to reliably detect it
The rosiest estimate	Backed out from the small early trial’s own numbers	About 1 point
The company’s own estimate	What Amgen itself assumed when originally planning the small early trial	About 1.3 points
The most realistic outside check	Backed out from the one real comparable trial that already happened (the competitor drug, ianalumab)	About 1.5 points

Why three numbers instead of one matters: an earlier version of this analysis used only the rosiest estimate, which made the whole prediction look far more confident than it should have. Using the more realistic, externally-checked number still supports a positive lean — it just means the safety margin is real, but smaller and more honest than first described.

Bull, base, and bear — what these words mean and the actual numbers

Because nobody can know the trial’s real result in advance, three different “what if” scenarios were built, using the more realistic effect-size estimate above:

- **Bull case** (“everything goes well”): imagine the true effect is a healthy 2.0 points. Under this assumption, the trial would pass with a p-value under 0.1% — an extremely comfortable, clear win.
- **Base case** (“the most likely middle-ground guess”): imagine the true effect is a more modest 1.4 points — smaller than what the small early trial showed, accounting for the fact that positive early results often shrink somewhat once tested in a much bigger, more spread-out trial (a well-known pattern in medical

research called **regression to the mean** — small early results tend to be a little lucky, and bigger follow-up tests bring the number back down closer to the true average). Under this assumption, the trial would still pass, with roughly a **0.8% chance** it's a fluke — a real, meaningful pass, though not an overwhelming one.

- **Bear case** (“things go poorly”): imagine the true effect shrinks a lot further, down to just 0.6 points. Under this assumption, the trial would **not** reach the normal bar for statistical significance — there'd be roughly a **1-in-4 chance** the whole thing is just noise, which is far too risky to call a real result.

Important honesty check: these three numbers are not measured predictions of what will actually happen — they're illustrative “what if” inputs, stress-tested to see how the trial's result would look under each assumption. Nobody knows today which of these (if any) is the truth. Based on all the evidence gathered, a rough personal judgment call was made about how likely each scenario is: about a **1-in-5 chance** of the bull case, a **1-in-2 chance** of the base case, and a **roughly 1-in-3 chance** of the bear case. This weighting is a considered opinion, not a scientifically computed probability — it's explicitly labeled as a judgment call everywhere it's used, so it can be challenged or adjusted.

So... why does the final answer lean positive?

Putting all of Day 1 and Day 2's evidence together, here is the actual reasoning, ranked from strongest to weakest:

1. **The math has real breathing room.** Because the big trial has so many patients, it doesn't need anywhere close to the small early trial's full 2.2-point improvement to count as a genuine success — even the more cautious, realistic “base case” estimate (1.4 points, meaningfully smaller than what was already shown) still passes comfortably.
2. **The blood test evidence is independent proof the drug is biologically doing something real** — separate from the symptom scores, which are known to be a noisier, less sensitive way to measure this disease (see Step 3 above). This makes “the drug doesn't actually work” a less likely explanation for any future miss.
3. **A very similar competing drug already proved this exact kind of trial can succeed in this disease** — real-world evidence, not just theory — and the current drug's own early results were already stronger than what carried that competitor through its own big trial.
4. **Both drugs sharing this drug's specific biological approach (blocking either side of the same signal) have, so far, cleared their own early testing goals** — a small but genuinely encouraging pattern.

The single strongest reason this could still fail

Being honest about the biggest real risk matters as much as the reasons for optimism. **The one real comparable example available — the competing drug's own jump from a small trial to a big one — only involved a modest increase in the size of the specific group being compared (about 2.9 times bigger). This drug's jump is meaningfully larger (about 6 times bigger).** *(Correction: an earlier version of this document compared the wrong numbers — total trial size instead of the specific group size that actually matters for the statistics — which made the gap look far more dramatic, 8.8× vs. 1.4×, than it really is. The honest comparison is 6× vs. 2.9×: still a real, meaningful difference, just not as extreme as first stated.)* There's no real precedent for what happens to a drug's effect when the jump in trial size is that much larger. If the true effect shrinks by more than what happened to that one comparable example, the result could land in the exact danger zone where several other past trials in this same disease have seen a real effect disappear into statistical noise. This is a specific, checkable risk — not a vague worry — and it's the fair, honest counterweight to the positive lean above.

The second, smaller trial (day-to-day symptoms)

This one gets a lighter, less mathematically detailed treatment by design. The takeaway: **lean positive here too, with slightly more confidence than the big trial**, because its own small early result (the 1.3-point ESSPRI improvement mentioned in Step 1) was the single cleanest, most convincing number in the entire early-trial program. The trade-off: this trial’s scoring tools (the ESSPRI questionnaire and the DASPRI diary) are both entirely based on what patients say they feel, with no doctor-measured or blood-test anchor at all — which historically makes this kind of measurement even more exposed to the placebo effect described in Step 3. This trial also hasn’t finished collecting its data yet, unlike the big trial, so its result will likely become known later.

5. What is the “reference deck,” and how are we using it?

Jefferies supplied a real, finished example of this exact kind of analysis — a deck one of their own analysts wrote predicting a completely different drug’s (a fatty liver disease treatment) trial result, before it was announced. It’s not the topic to copy — it’s a **style and rigor template**.

What we borrowed from it: - Every slide making an actual point/conclusion in its title, not a vague label. - Building an explicit “what changed between the small trial and the big trial” table with real numbers, rather than hand-waving about it. - Presenting three scenarios (bull/base/bear) with real numbers attached to each, rather than just saying “it could go well or badly.” - Cross-checking assumptions against other, similar real-world trials rather than inventing numbers out of thin air. - Naming one specific “here’s the strongest reason we could be wrong” rather than a vague list of risks.

What we added or did differently: - Testing three different “how much natural noise to expect” assumptions instead of picking one — the reference deck used a single number for its size assumption. - Being explicit about which numbers this project is *not* able to calculate (like an overall single percentage chance of success) because the necessary confidential company information isn’t public — the reference deck’s original author had more company-disclosed information to work with for its own drug. - Treating blood-test/biological evidence as a separate check on *why* a future miss might happen (bad ruler vs. bad drug), which wasn’t as necessary for the reference deck’s drug since its own measurement tool was already a lab test.

6. What’s left to do

The actual science and statistics work described above is essentially complete. Two small, genuine research loose ends remain (writing down the biological-rationale explanation as its own saved file, and pulling one specific missing data point about the second trial’s diary-based measurement), but nothing close to a full new research cycle. The large remaining task is building the actual PowerPoint presentation from everything already gathered — turning the research above into slides, tables, and charts — which has not yet started.

7. Does this cover what Jefferies actually asked for?

What they asked for	Where it stands
A clear, explicit “will it pass or fail” call	✔ Done — see Day 2 above
Explanation of the biology and why the drug should work	✔ Covered here, though it should also live as its own separate file
Analysis of the early small trial	✔ Done in detail
Explanation of what’s unique about the two big trials’ designs, and the statistics involved	✔ Done

What they asked for

Lessons learned from other companies' trials

Sourced, traceable numbers throughout

An actual PowerPoint file

Where it stands

✓ Done, double-checked for accuracy

✓ Done — every number traces back to a real source, documented in a master evidence file

✗ Not started yet

Bottom line: the thinking and the evidence are done and have already been independently checked once for mistakes. What's left is turning it into slides.